6 Fig. 6.1 shows the circuit diagram for a flashlight (torch).

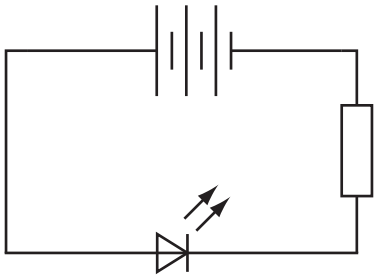

Fig. 6.1

The electromotive force (e.m.f.) of the battery is 4.5 V. The circuit contains a 60 Ω fixed resistor. The current in the light-emitting diode (LED) is 0.020 A.

(a) Calculate the potential difference (p.d.) across the LED.

p.d. = ... [2]

(b) Explain why the LED does **not** light up if the battery is reversed.

.....

..... [1]

(c) The chemical energy stored in the battery is 1050 J.

Show that the flashlight operates for approximately 3 h.

[2]

(d) Calculate the total charge that flows through the LED in 3600 s.

charge = ... [2]

[Total: 7]

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